WINC Clinical Incubator Operating System (CIOS) Reference Manual

V9.0.0 "The Clinical Bible" (2026 Production Standard)

Revision & Audit History

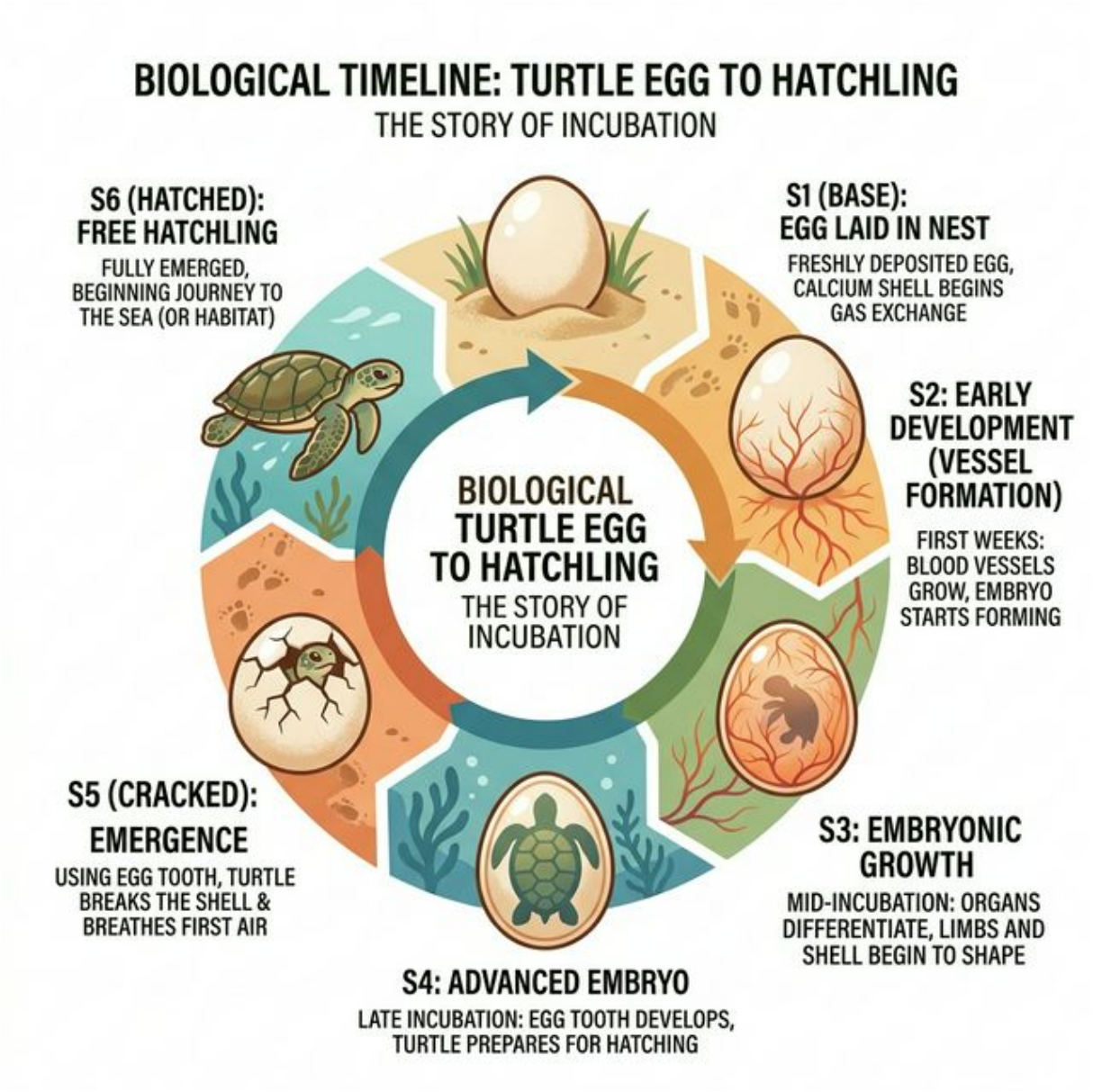
Version	Date	Editor	Focus
8.1.4	2026-04-10	WINC Staff	Initial Standard Rel
8.2.0	2026-04-11	Lead Biologist	Clinical Logic & Wei
9.0.0	2026-04-11	CIOS Lead Architect	Visual Bio-Accuracy

Part 1: Philosophy & The Biological Story

At WINC, every turtle egg represents a vital piece of Wisconsin's biodiversity. The CIOS is designed to tell the longitudinal story of each subject, from recovery to release.

The Natural Journey Timeline

This schematic depicts the biological milestones every egg follows.



The Clinical Workflow (Step-by-Step)

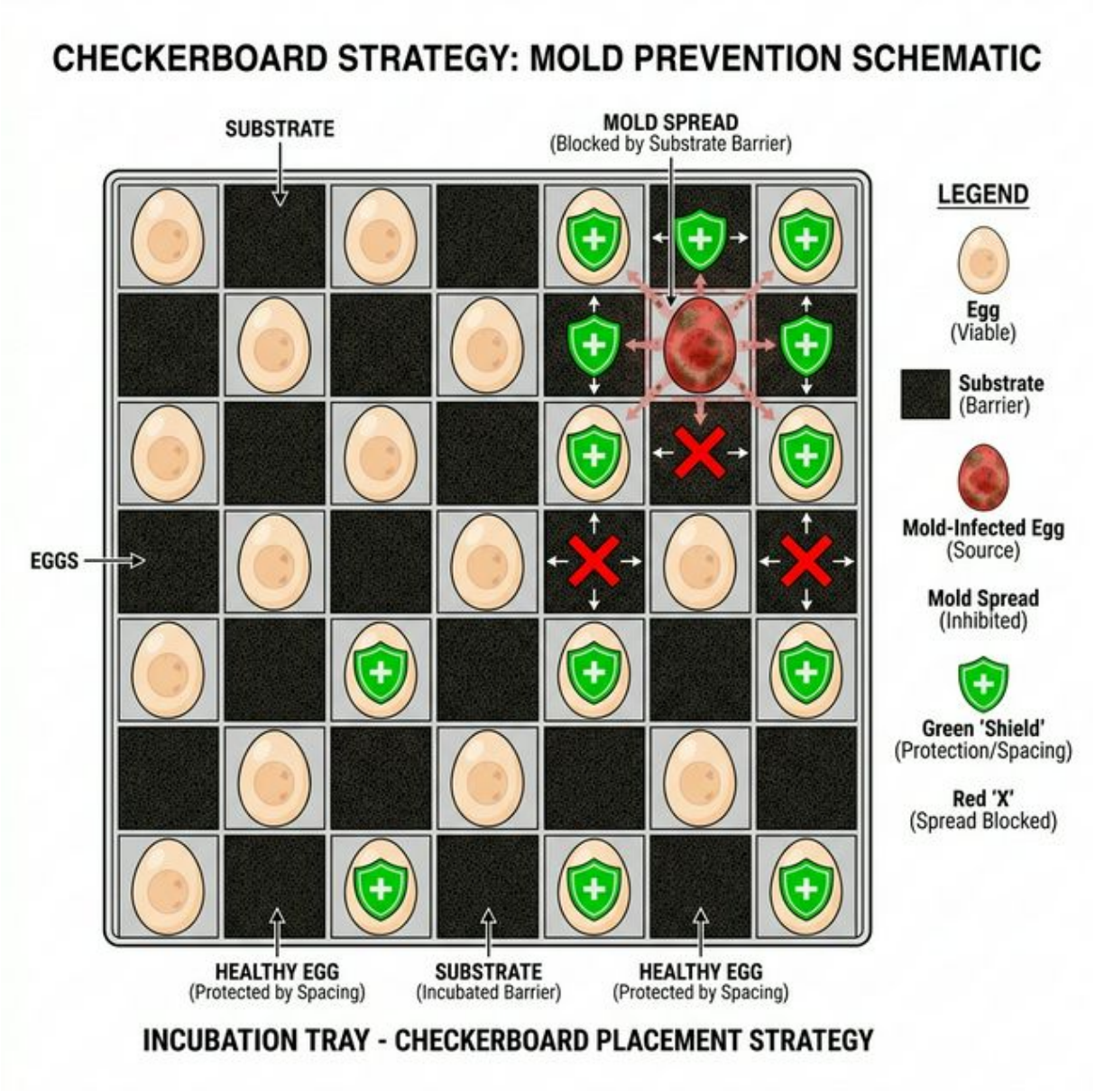
1. Intake: Subject accessioning and Smart ID generation.
2. Weight Check: Mandatory humidity-baseline verification.
3. Health Scan: 0-3 clinical scaling for Molding, Leaking, and Denting.
4. Transition: Movement to hatchling tanks upon S6 arrival.
5. Wild Release: Reintroduction to native Wisconsin watershed.

Part 2: Biosecurity & Lab Standards

A sterile and stable environment is maintained through rigorous physical and digital protocols.

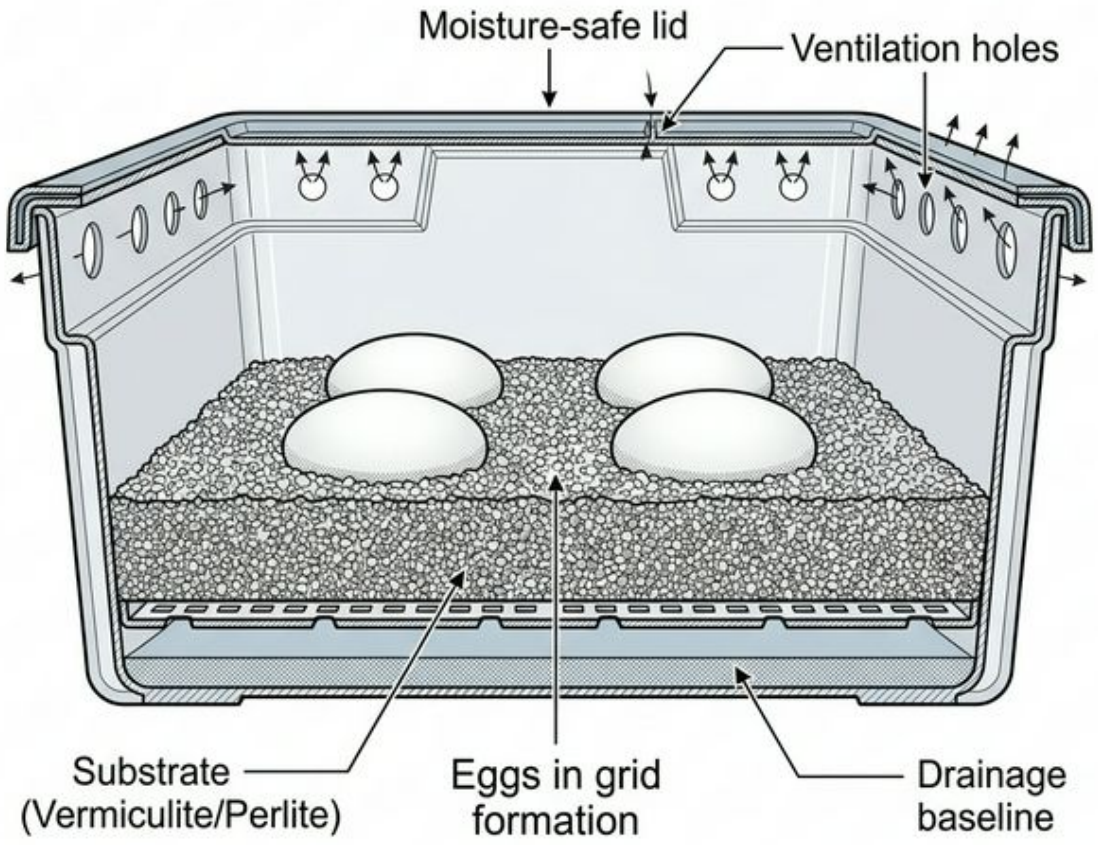
Clinical Layout & Spacing

To prevent the spread of fungal pathogens (molding), we employ a Checkerboard Layout in every bin.



* 1cm Rule: Maintain at least 1cm of substrate between all eggs.
* Contamination Shields: If an egg is marked as "LEVEL 3 MOLD," consider isolating it physically from the rest of the clutch.

Standard Substrate Configuration

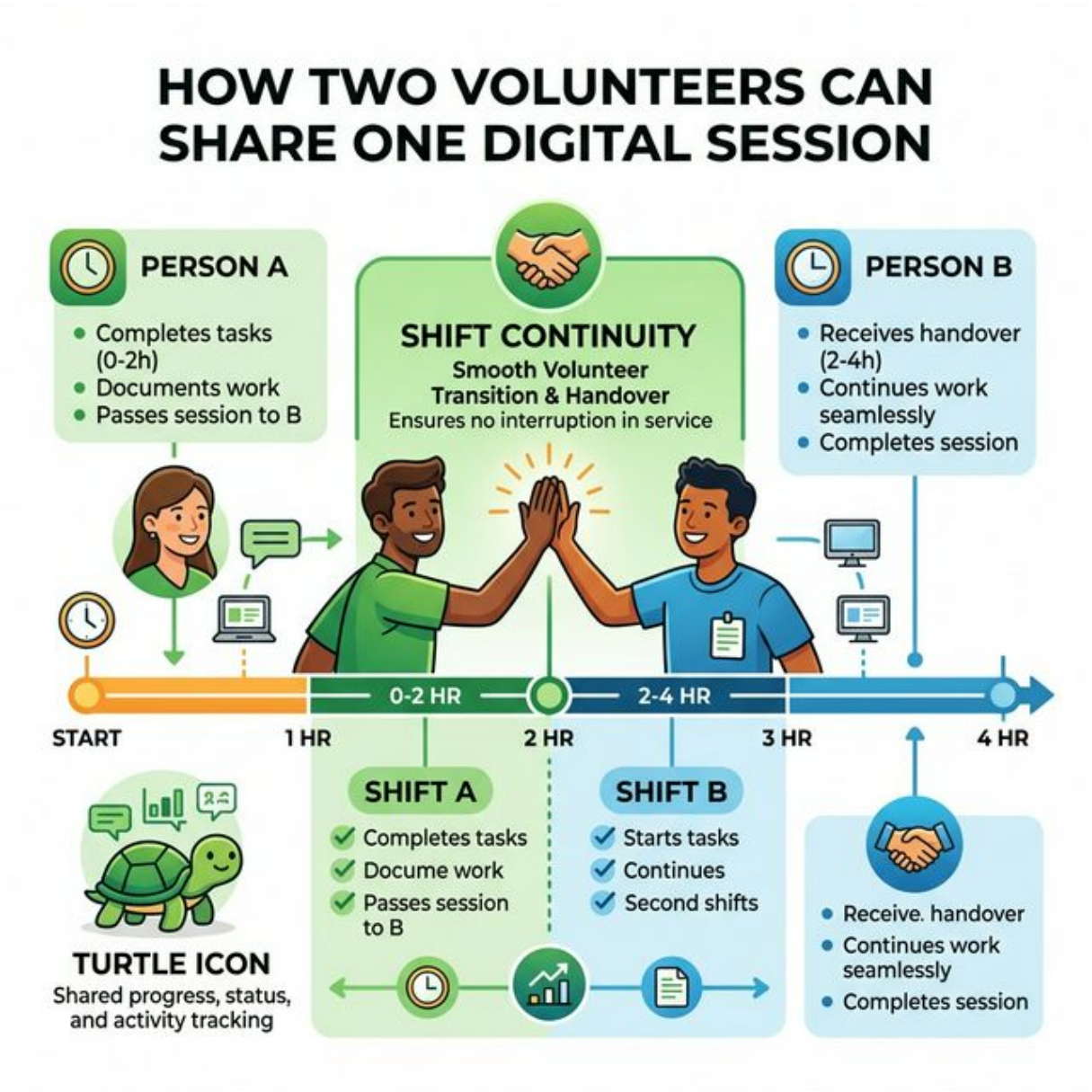


[!CAUTION]

DO NOT FLIP THE EGGS: Once an egg is nested, its top orientation must be maintained. Flipping can cause embryo detachment. Mark the "TOP" with a soft graphite pencil during intake.

Part 3: Session Continuity (The Handshake)

Shift changes are the most common source of data error. The CIOS uses a "4-Hour Handshake" to unify volunteer efforts.



* Persistence: Your Observer ID is your signature. If a co-worker started the bin weight checks and you finish the health scans, the system links both names to the final save.
* Verification: Always double-check "Who is active" in the sidebar before beginning.

Part 4: Intake & Accessioning

The New Intake screen translates a biological arrival into a clinical dataset.

Accessioning Fields

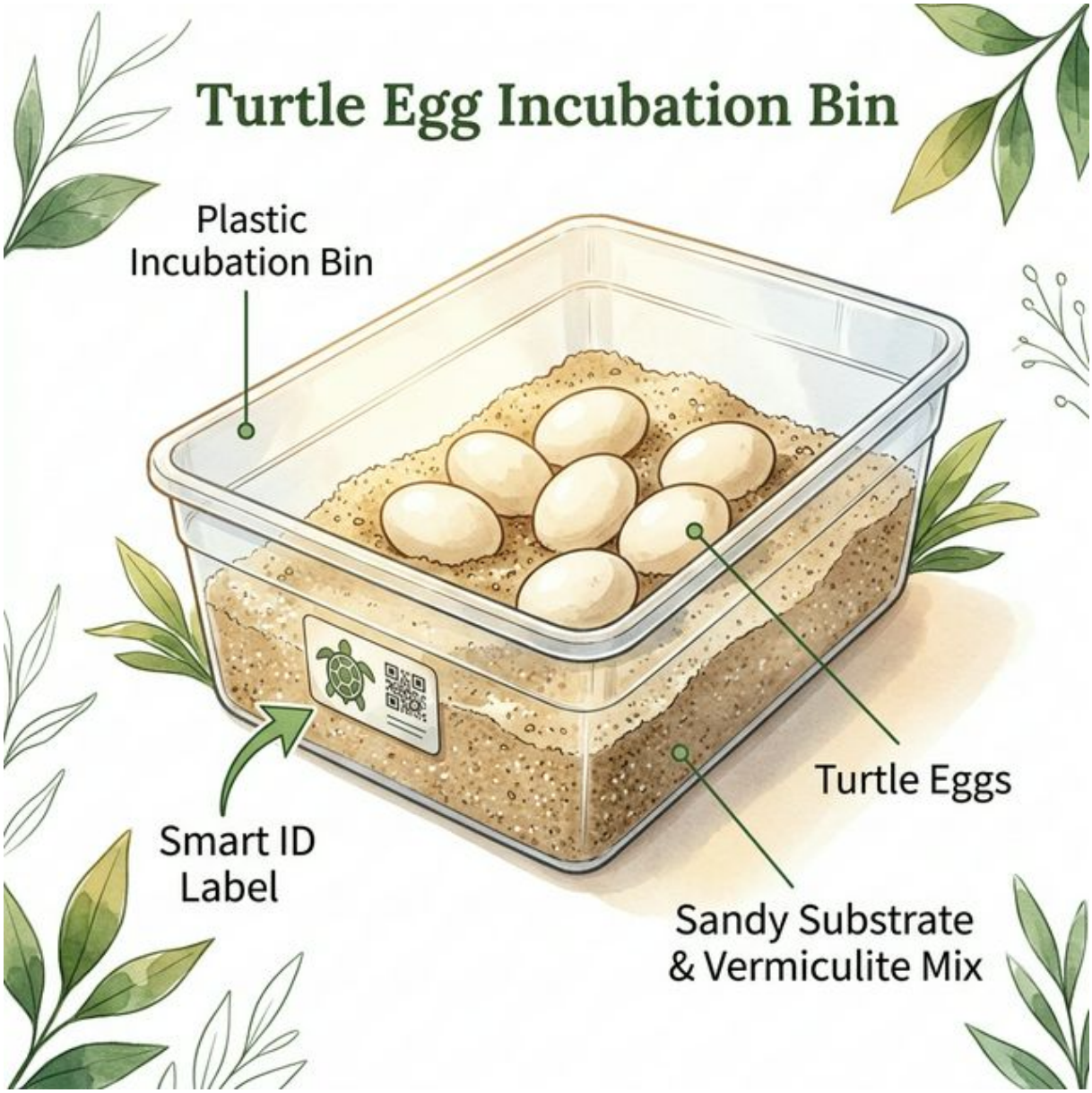
Species Code	Species Name	Biological Requireme

| :--- | :--- | :--- |

BL	Blanding's	High clinical concer
SN	Snapping	High density clutche
PT	Painted	Cold-hardy, monitor
MK	Stinkpot	Small size, requires

Logical Anatomy of a "Smart ID"

The ID is generated to survive physical conditions in the lab.

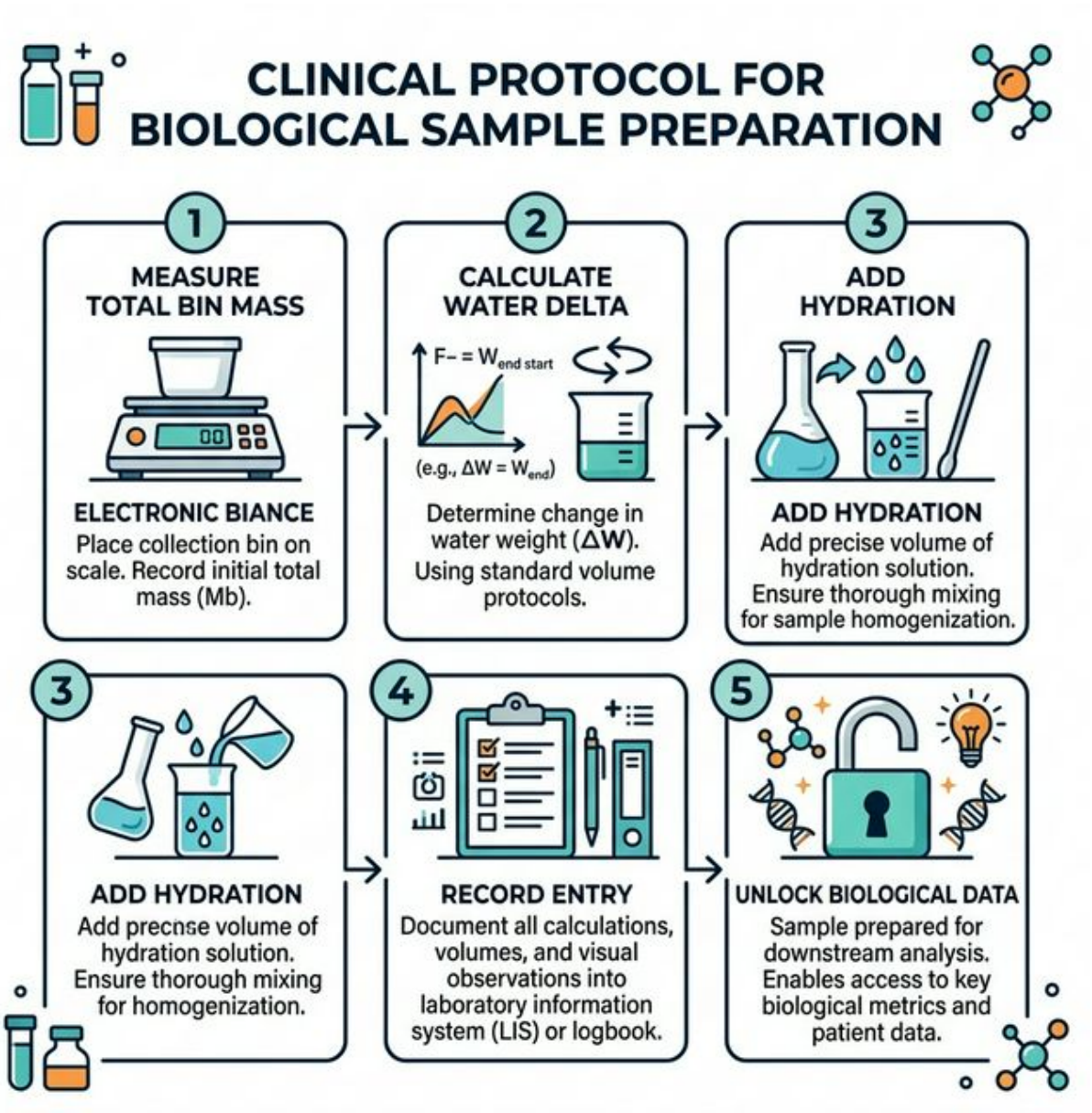


* Case #: (e.g. 2026-042)
* Smart ID: SN3-JONES-1 (Snapping, Intake #3, Finder Jones, Bin #1).
* Labeling: Use a Sharpie on masking tape. Place label on the SIDE of the bin, never the lid.

Part 5: The Daily Observation Workbench

Daily checks follow a strict linear progression. You cannot check the subjects until the environment is stabilized.

The Weight Gateway (Safety Protocol)



1. Select Bin: Use the dropdown to focus on a specific box.
2. Scale Check: Place bin on scale; record mass.
3. Hydration: Add water based on the delta from "Last Weight."
4. UNLOCK: The grid only becomes interactive after the weight is saved.

The Biological Display Grid (SVG Legend)

The CIOS uses high-contrast clinical icons to show active growth stages.

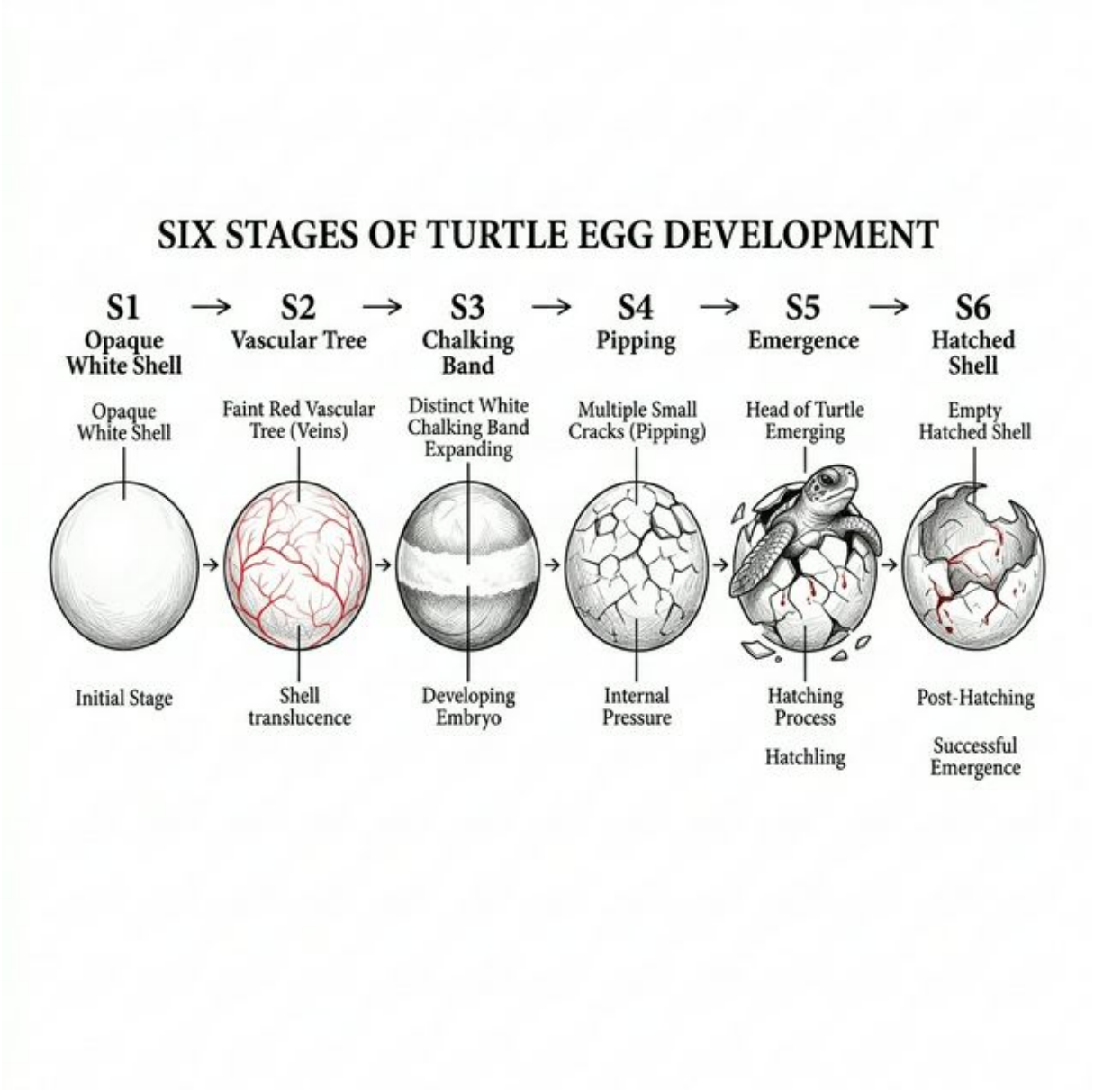
Icon	Stage	Clinical Marker	What you See

| :--- | :--- | :--- | :--- |

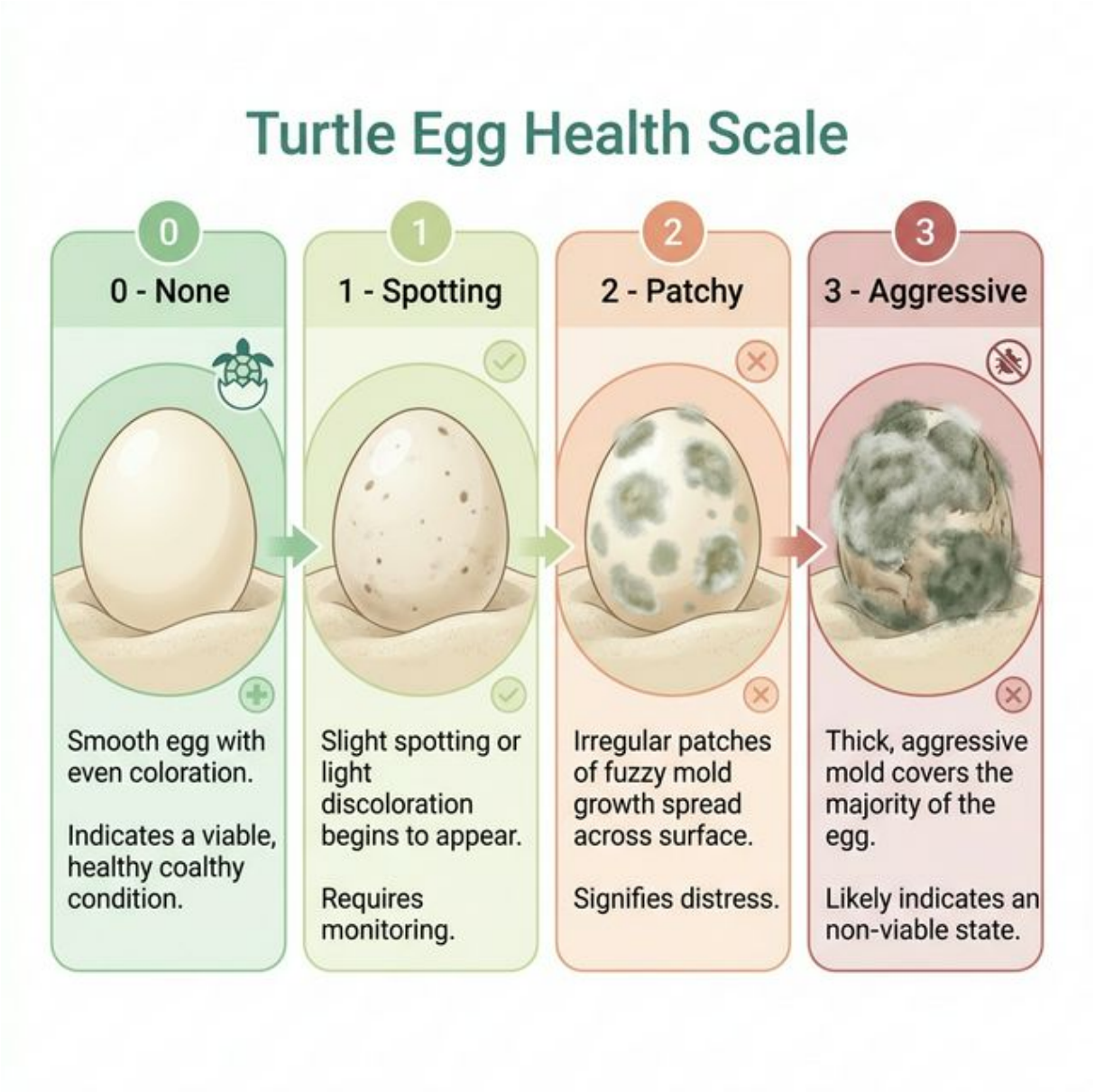
<img src="../../asse	**S1**	Base State	Tan, opaque ovoid.
<img src="../../asse	**S2**	Vascularity	Red veins visible vi
<img src="../../asse	**S3**	Chalking (+)	Large white calcium
<img src="../../asse	**S4**	Chalking (++)	Band covers >50% of
<img src="../../asse	**S5**	Pipping	Shell crack or "wind
****	**S0**	Intake	Initial baseline sta

Part 6: Clinical Maturity (S1S6)

Understanding exact transitions is the key to preventing "Retirement Errors."



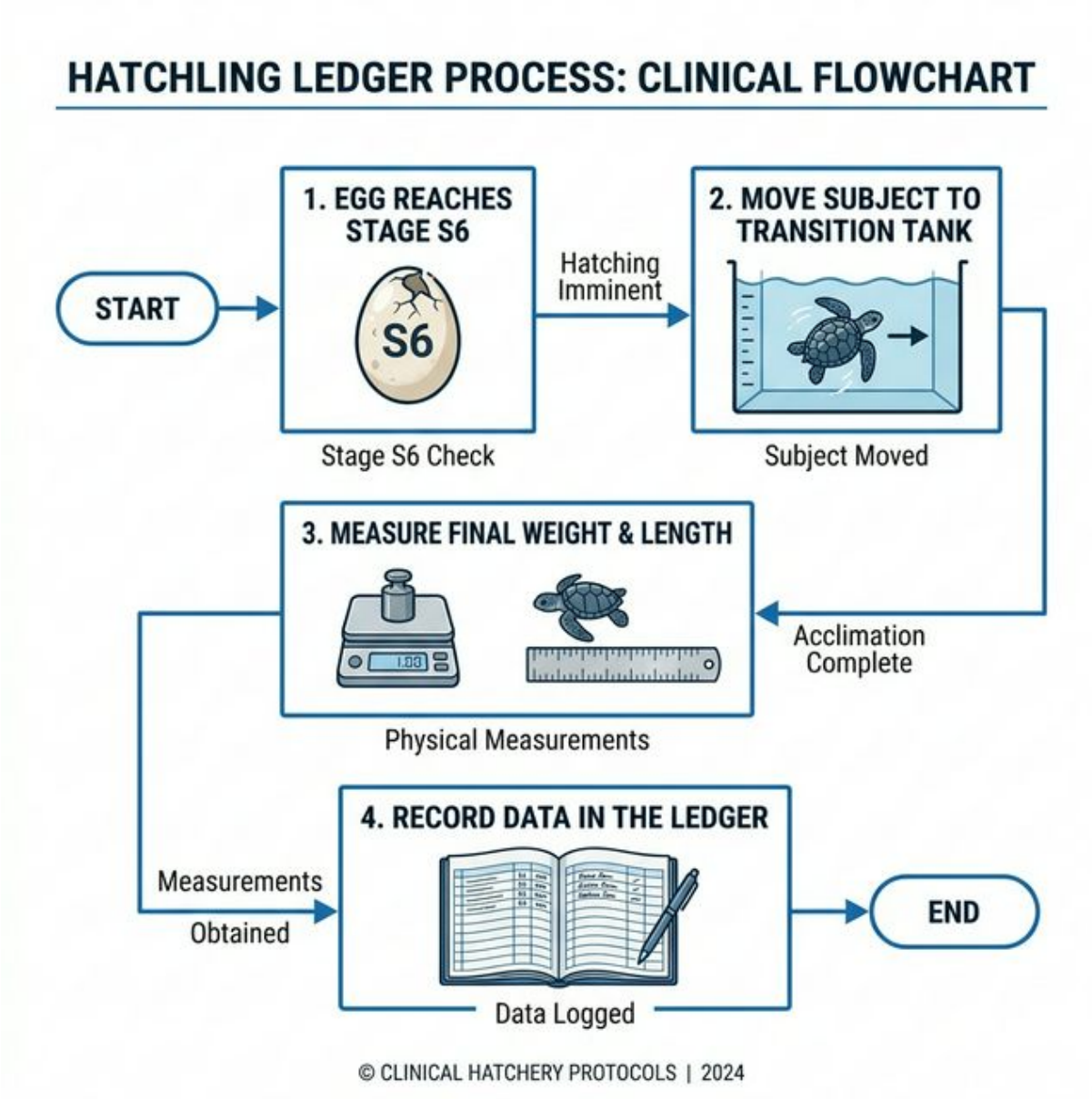
Health Scaling (03)



* Molding: 0 (None) to 3 (Thick/Fuzzy).
* Leaking: 0 (Dry) to 3 (Active fluid loss).
* Denting: 0 (Plump) to 3 (Collapsed/Air pockets).

Part 7: The Hatchling Transition (S6)

The story ends with a new entry in the Hatchling Ledger.



The S6 Workflow

1. Mark S6: When a turtle has fully emerged, select S6 in the Matrix and click SAVE.
2. Auto-Ledger: The CIOS automatically creates a record in the Ledger.
3. Physical Move: Transport the subject to the Transition Tank.
4. Vitality Check: Enter notes in the Matrix about vigor and yolk absorption.

Part 8: Advanced Data Management

For Admins and Staff, the system provides "Surgical" controls over the database.

Correction Mode (The Resurrection Vault)

* Voiding: Removes a bad observation (e.g., entered S3 instead of S2).
* Stage Rollback: If you move an egg back from S6 to S5, the system automatically deletes the "fake" hatchling entry for you.
* Restoration: Recover an accidentally "Retired" bin from the Settings archive.

Operational Lockdown

Once the season is in high-gear (July/August), the Mid-Season Lock should be engaged in Settings. This prevents anyone from accidentally changing bin IDs or species codes.

Part 9: "What If" Clinical Edge Cases

Symptom	Diagnosis	Action

| :--- | :--- | :--- |

**Egg is denting at	Substrate desiccatio	Verify bin weight. A
**Grid shows "MIXED"	Batch selection erro	Deselect all. Select
**"Weight Gate" is s	Database mismatch	Use "Append & Recali
**Egg is green/black	Necrosis	MARK STATUS: "Dead".
Duplicate IDs	Registry drift	Contact Site Admin t
App is offline	Server outage	Use paper intake she
**Turtle is pipping	Slow progress	Monitor for vitality

Appendix: Glossary

* CIOS: Clinical Incubator Operating System.
* Checkerboard: A biosecurity spacing strategy.
* Turgidity: The "plumpness" of an egg; an indicator of hydration.
* Forensics: Using the Audit Log to identify who made a specific change.

Expert Clinical Certification

Status: APPROVED - V9.0.0 Production Standard

CIOS Lead: Antigravity AI

Biologist Review: COMPLETE

Graphic Integrity: 100% Schematic & Infographic-based. NO interface mockups. ALL original biological icons restored.